Abdelrahman Elmogy

Data & Automation Engineer

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Cairo, Egypt · Open to Remote & Relocation Worldwide

PROFESSIONAL SUMMARY

Data & Automation Engineer with 3+ years of experience turning business and operational problems into practical technical solutions. Understands the process, identifies the root cause, and designs the right approach before building it, combining business intelligence, Python engineering, system integration, workflow automation, and AI to deliver measurable results. Built and deployed an AI-powered operational solution integrated with SQL Server for a leading debt collection operation, connected systems across SQL Server, APIs, and Power BI, and automated multi-step workflows that eliminated hours of manual work each week.

Gulf Telecom	7+ Systems	8 hrs/week	10+
Enterprise Clients	Automated	Reporting Time Saved	High-Impact Projects

TECHNICAL SKILLS

Data & BI	Power BI · DAX · Power Query · SQL Server · Excel · KPI Design · Dashboard Architecture
Data Engineering	Python · Pandas · NumPy · ETL · Data Pipelines · Database Design · SQLAlchemy · pyodbc
System Integration	REST APIs · API Integration · Webhooks · Data Integration · SQL Databases · Real-Time Data Flows
Automation & AI	Workflow Automation · Business Process Automation · AI Agents · Conversational AI · AI-Powered Automation · Task Scheduling
Development	Flask · FastAPI · Selenium · BeautifulSoup · openpyxl · Git
Infrastructure	Windows Server · Linux · cPanel · System Migration

PROFESSIONAL EXPERIENCE

Data & Automation Engineer · Maharah International for Debt Collection

Jul 2024 – Sep 2026

Cairo, Egypt

AI-Powered Operational Automation

► Analysed call-handling bottlenecks in a high-volume outbound operation, then designed and built an AI-powered operational solution integrated with SQL Server, reading and writing live business data in real time. First solution of its kind at the company, with outcomes tracked to refine performance.

Business Intelligence & Reporting

► Analysed reporting gaps across departments and built live Power BI dashboards connected to production databases, giving management real-time KPI visibility. Eliminated a weekly manual reporting cycle of ~8 hours and enabled same-day operational decisions.

Automation & Python Engineering

► Identified recurring data-entry errors through workflow analysis, then automated core processes using Python, cutting manual effort by 60% and eliminating discrepancies that had affected financial reporting.

► Studied campaign dispatch patterns and built Campaign AutoFlow, a data and integration solution that pulls segmented target lists from SQL Server and dispatches IVR, WhatsApp, and email campaigns simultaneously through their respective APIs, writing outcomes back automatically. Cut execution time by 60%.

Operational Analytics & Automation Strategy

► Led performance analysis of IVR, WhatsApp, and email campaigns using Power BI and Python, translating findings into optimisation and automation strategies adopted across the department.

System Administrator & Data Analyst · **Aug 2023 – Jun 2024**

Royal Canadian School

New Administrative Capital, Egypt

► Analysed fragmented record-keeping and built data management systems in Excel and SQL Server covering 500+ student and staff records.
► Studied the finance department's forecasting workflow and designed a live Power BI budget tracking dashboard, enabling accurate real-time financial forecasting for the first time.
► Managed full IT infrastructure, including networks, servers, and workstations, maintaining high uptime across multiple departments.

Freelance Python Developer · Independent · **Oct 2023 – Present**

Remote

► Analysed client workflows across multiple industries and built end-to-end Python automation systems, including web scraping pipelines, WhatsApp bots, and email campaign engines, reducing client workloads by up to 40% per engagement.
► Studied reporting needs for clients in finance, retail, and services, then developed custom data analysis solutions that replaced manual spreadsheets with automated dashboards for faster decision-making.
► Assessed operational gaps and designed business automation tools, including invoice processors, reconciliation scripts, and scheduled reporting systems, integrating SQL databases and REST APIs for reliable client solutions.

SELECTED PROJECTS

01 **Maharah AI Collection Agent** · *Conversational AI · Python · SQL Server · Real-Time API*
Analysed call-handling data across daily operations for leading Gulf telecom operators to identify friction points in live conversations. Built an AI-powered engine on SQL Server that reads account data in real time and adapts each interaction automatically. First of its kind at Maharah.

02 **Fraud Detection Model** · *Python · Scikit-learn · Pandas · Machine Learning*
Studied transaction-level data for device-signature clusters, timing anomalies, and usage outliers to define fraud risk indicators. Developed the detection model that scores and ranks transactions into a prioritised risk queue, replacing manual review.

03 **Cross Channel Campaign Dispatch Engine** · *Python · SQL Server · Multi-Channel Automation*
Reviewed campaign dispatch data, then engineered a data and integration solution pulling target lists from SQL Server and firing IVR, WhatsApp, and email simultaneously, with outcomes written back automatically. Cut execution time by 60%.

04 **Maharah Collection Performance Tracker** · *Power BI · DAX · SQL Server · Real-Time Dashboards*
Mapped reporting requirements across Maharah and Royal Canadian School, then built live dashboards tracking call centre performance, debt recovery, and financial budgets.

05 **Automated Commission Calculation Engine** · *Python · Excel Automation · Finance*
Diagnosed a slow, error-prone manual Excel process as the root cause of recurring payroll mistakes. Engineered an automated commission system that runs every cycle on its own, removing calculation errors entirely.

06 **TalkHub, a WhatsApp Business Automation System** · *Python · WhatsApp API · Messaging Automation*
Scoped a client's messaging requirements across support, sales, and broadcast use cases. Built a full-featured WhatsApp platform supporting two-way messaging, automated replies, broadcast campaigns, and complete conversation logging.

07 **LegalDesk, a Case and Document Management Platform** · *Python · SQL · Document Automation*
Mapped a legal client's intake and documentation workflow end-to-end to identify manual bottlenecks. Built the platform to run it, covering client intake, contract automation, case tracking, and document generation.

08 **PeopleOps, an HR Operations Platform** · *Python · SQL · Process Automation*
Audited workflow gaps across employee records, attendance, and payroll to scope the system requirements. Engineered a connected platform covering attendance tracking, recruitment, employee records, and payroll integration.

EDUCATION

Bachelor of Communication Engineering

Mansoura University, Egypt · 2018–2023

LANGUAGES

Arabic Native, **English** Prof, **Italian** Elem

AVAILABILITY

Available Immediately

Open to Remote & Relocation

On-site / Hybrid / Remote

CERTIFICATIONS

Data Analysis Professional, *Udacity 2023*
Data Analysis Challenger, *Udacity 2022*
Excel + Power BI Diploma, *Planet of Skills 2023*
Python & SQL Advanced, *HackerRank 2022*
Cyber Ops Associate, *NTI 2022*
CCNA Network Associate, *ITI 2021*
PL-300 Power BI Analyst, *Microsoft (In Progress)*